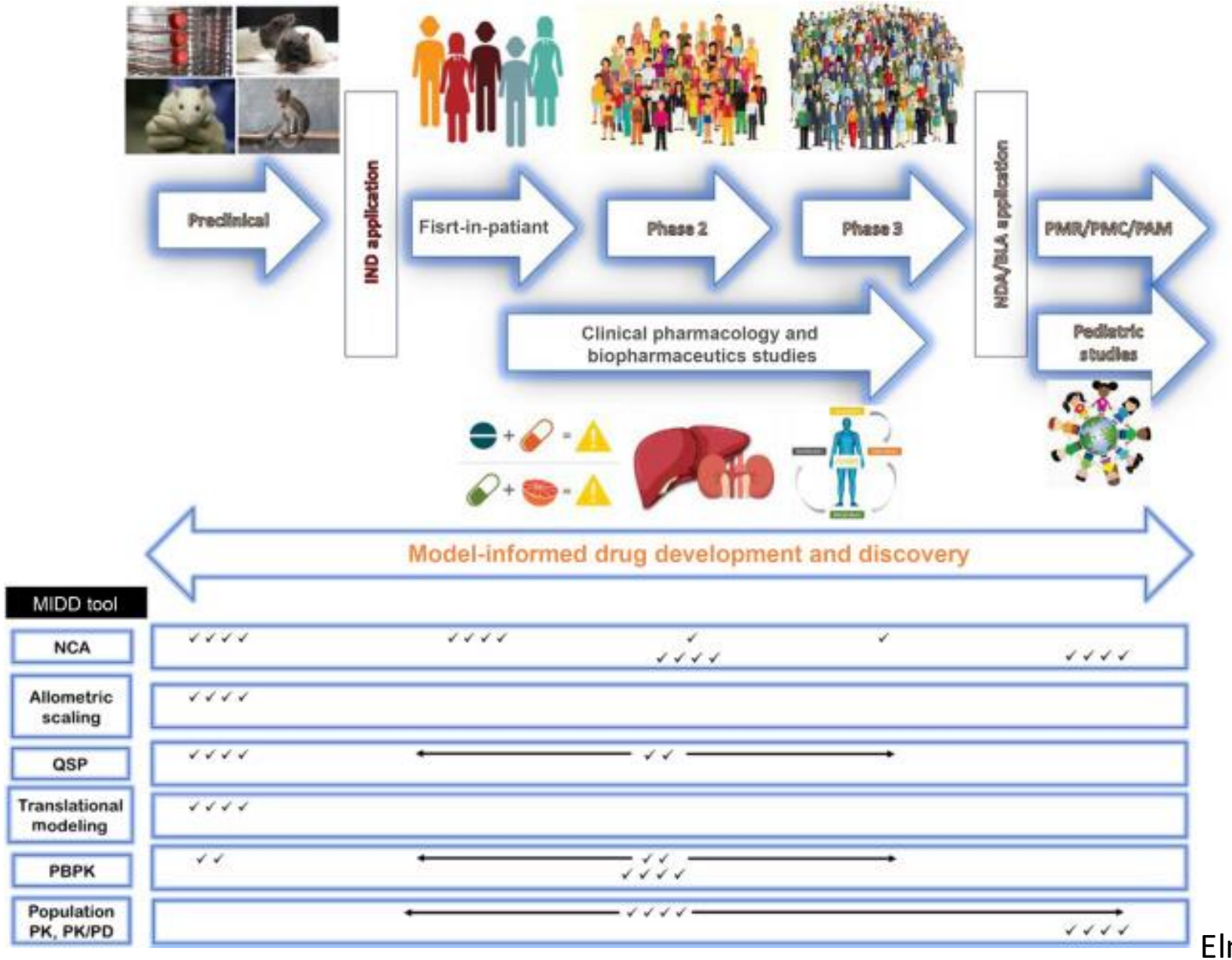
System dynamics workshop: the vaccine

Adam Darwich

Modelling Systems

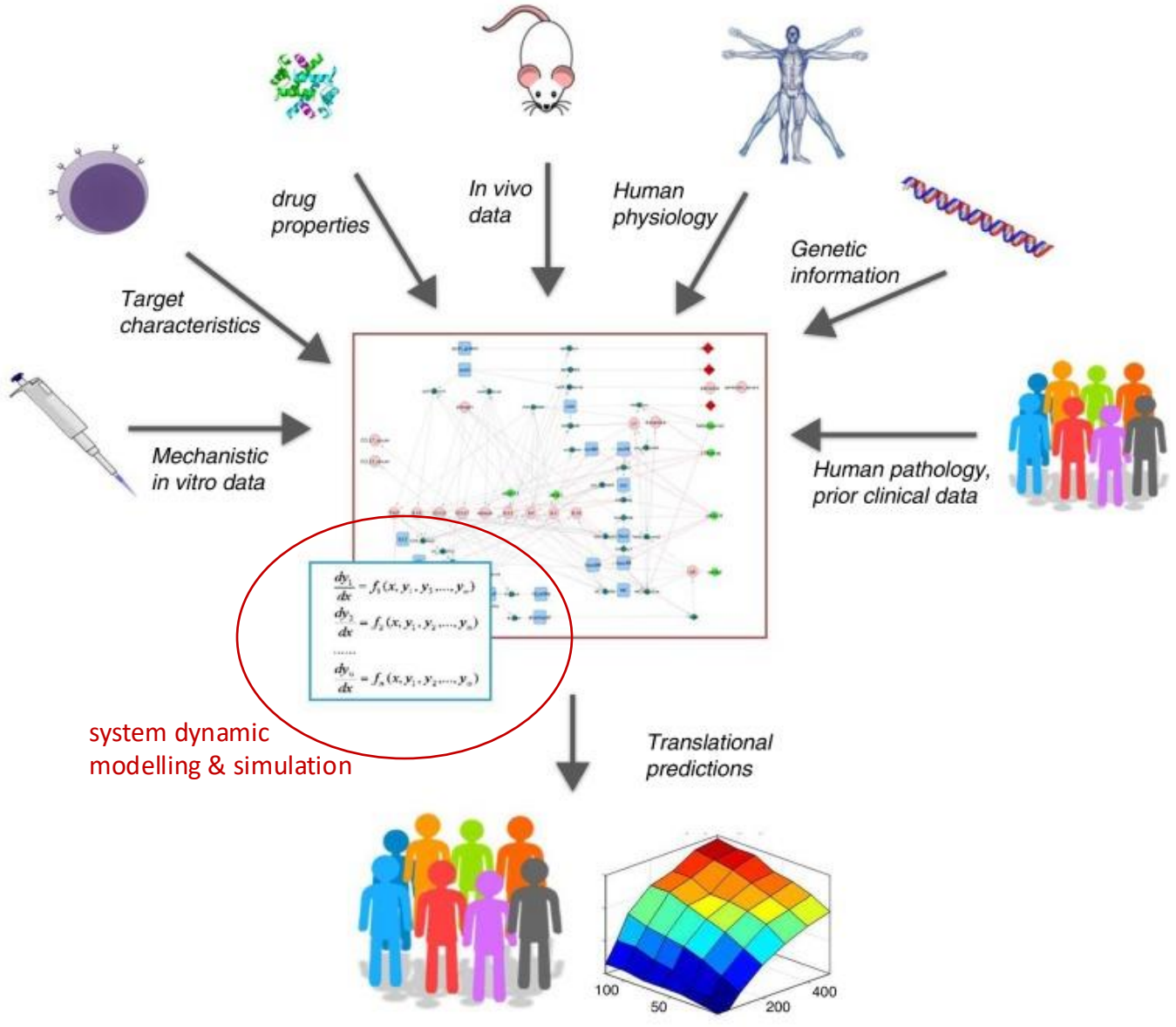
CM2014 Spring 2025

Modelling in drug and vaccine research & development



Elmeliegy & Ghobrial 2021

System dynamic modelling: Quantitative Systems Pharmacology (QSP)



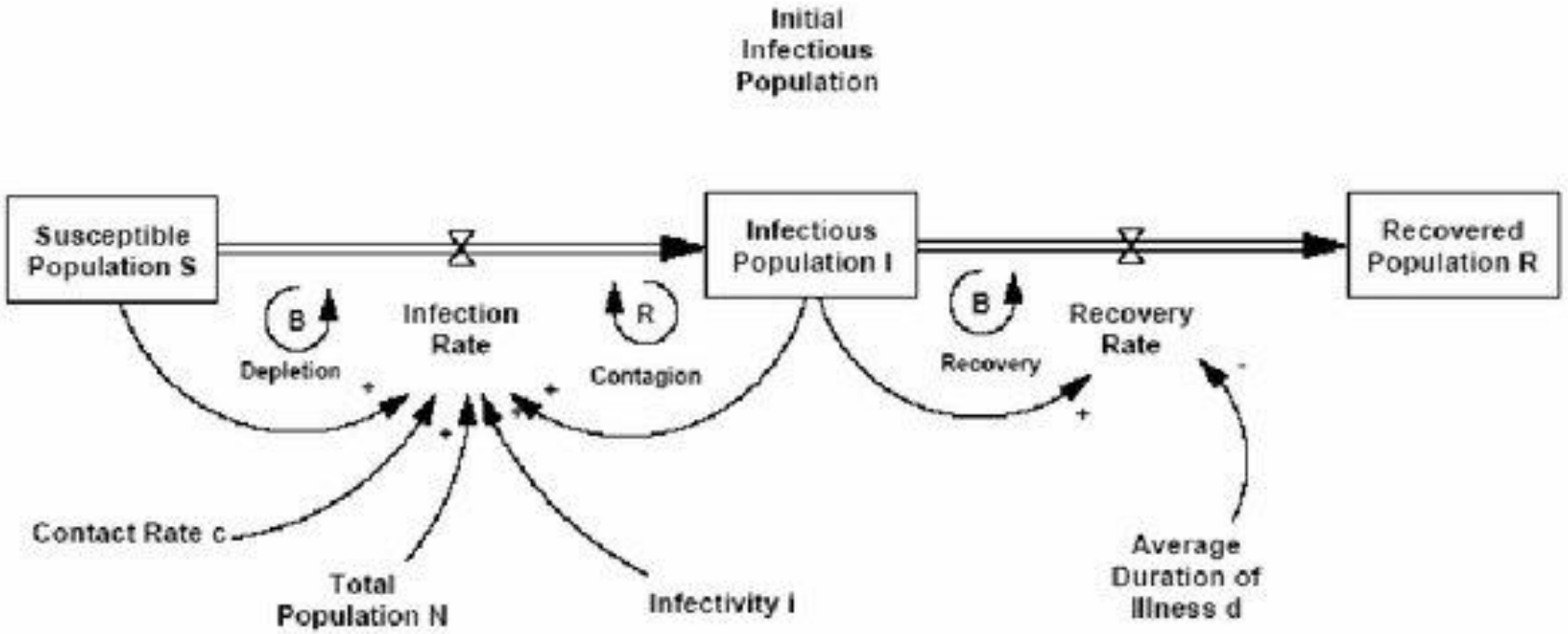
Drug Discovery Today: Technologies

Gadkar *et al*. 2016

The workshop

Aim and objectives:

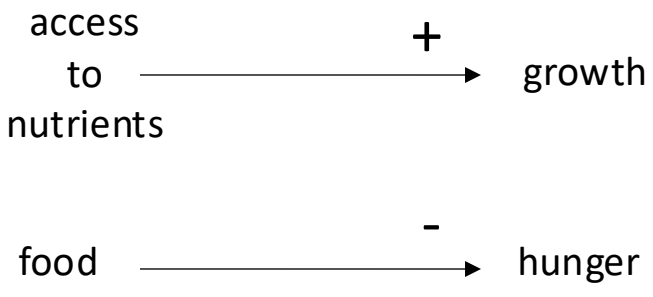
- Training on development of conceptual models based on problem description.
- Reflect on real-world use cases and how models can integrate large amounts of information.



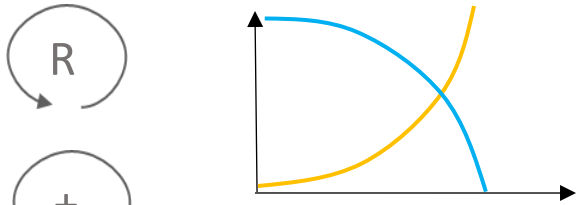
Saleh & Davidsen 2007

Some basic nomenclature of causal loop diagrams and stock-and-flows models

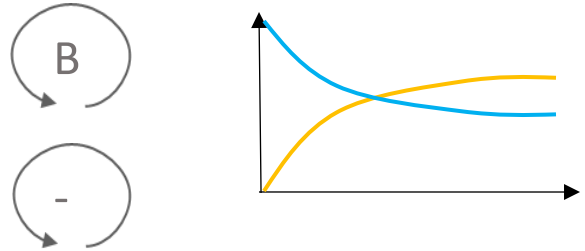
Variables, causal links, polarity



reinforcing feedback loop (R)



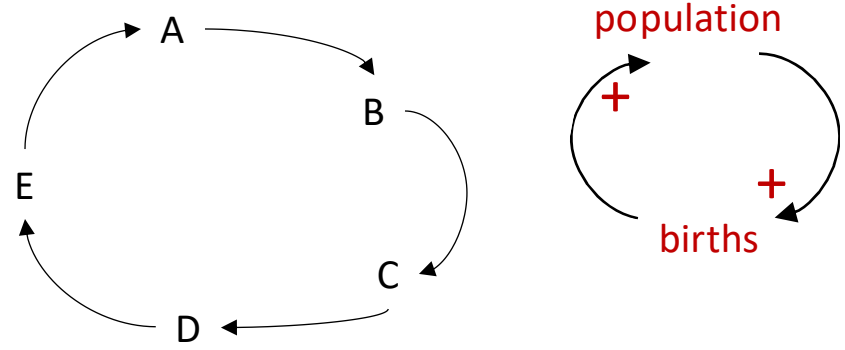
balancing feedback loop (B)



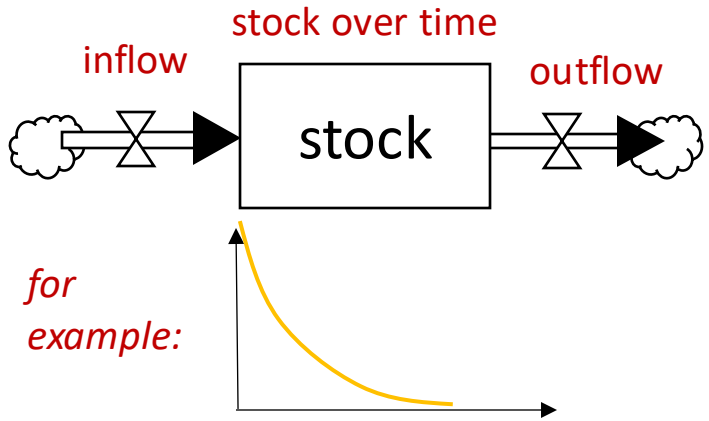
delay

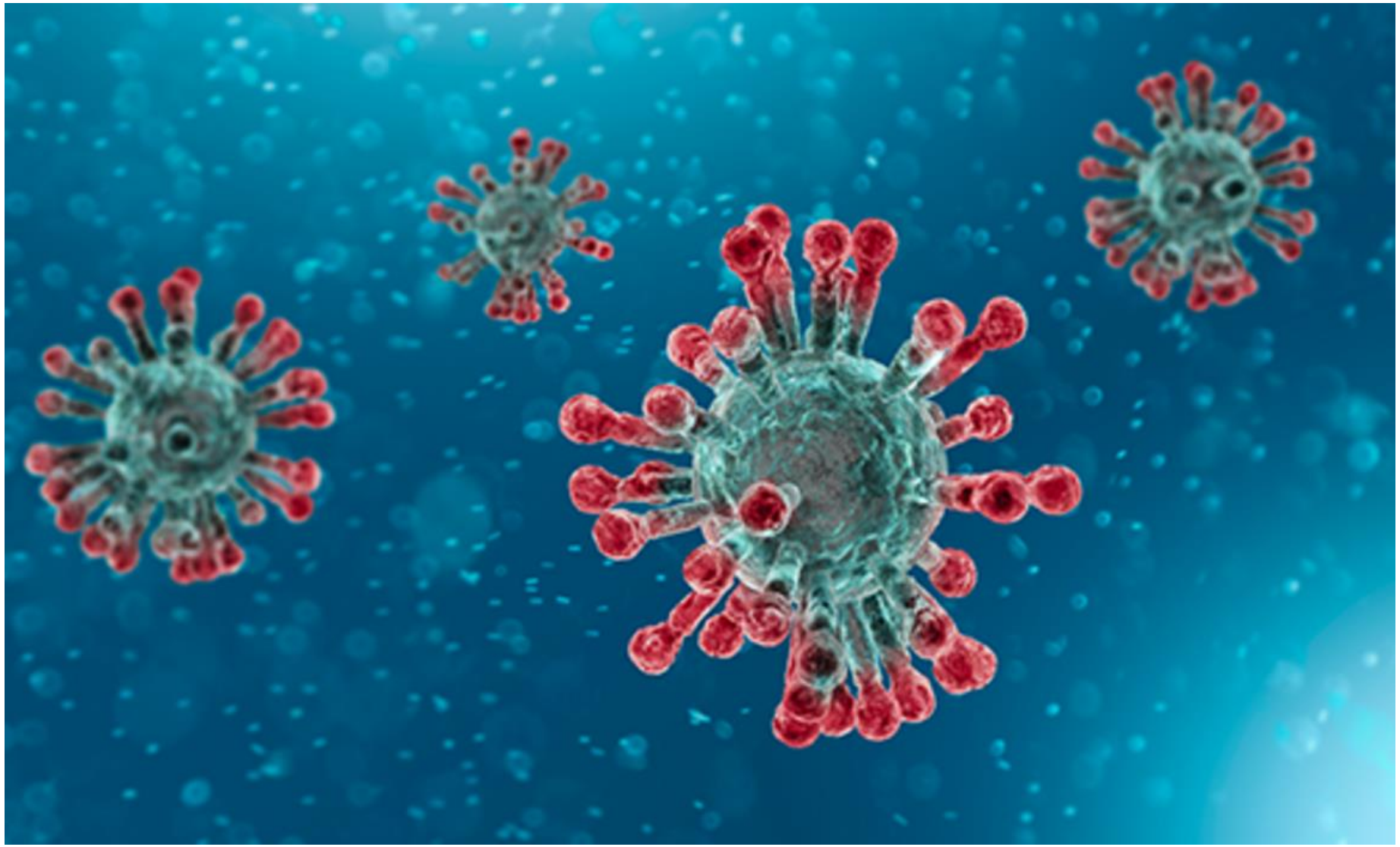


causal loops

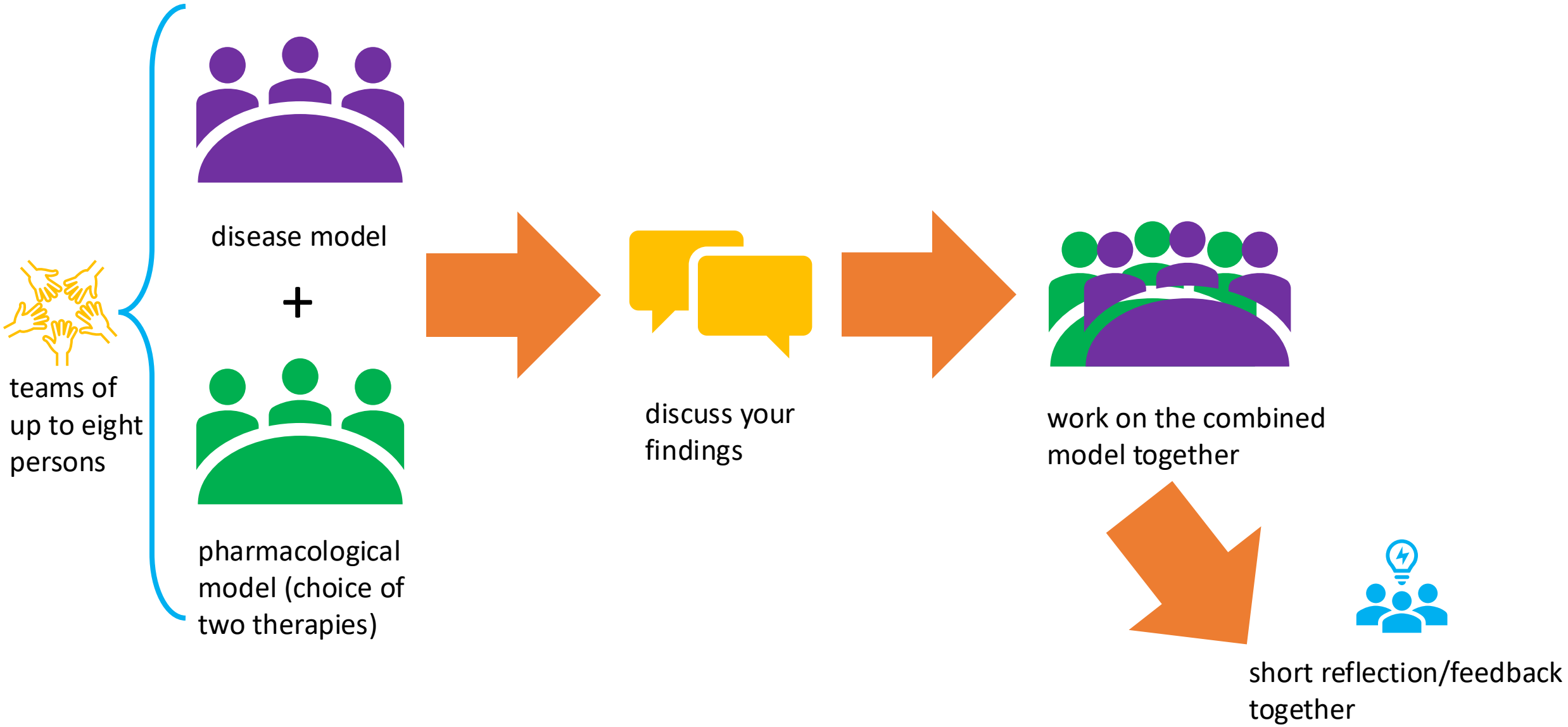


Stocks, flows, sources and sinks








disease model
+
pharmacological model (choice of two therapies)
teams of up to eight persons
discuss your findings
work on the combined model together
short reflection/feedback together

Schedule

13:30 – 14:00	Disease / Pharmacological model
14:00 – 14:15	Break
14:15 – 14:45	Combined model
14:45 – 15:00	Short reflection / feedback together

Model A: disease model

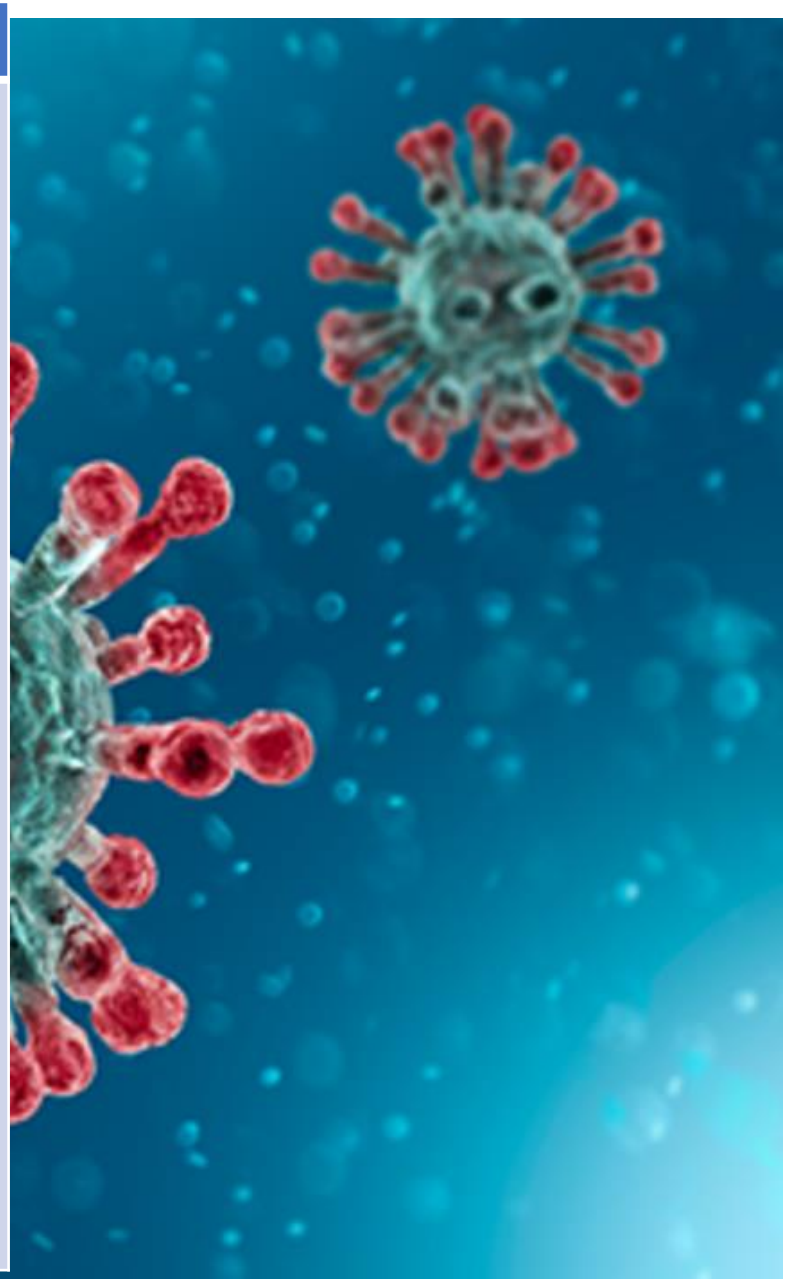
- **Problem statement**: Considering the mechanism of SARS-CoV-2, what are the potential drug/vaccine targets?
- **Task**: Based on the available information, develop a conceptual SARS-CoV-2 disease model (causal loop diagram/stocks-and-flows).
- **Question 1:** Given the structure of the model, what can we say about the above question?
- **Question 2:** Does the model explain the emergence of severe disease?
- **Question 3:** How and why would we want to translate this into a quantitative simulation model?

Model A: Disease mechanism - immuno-response model

Data Table. Disease mechanism of immuno-response.

- SARS-CoV-2 infection leads to increase in systemic blood viral load.
- Systemic blood viral load leads to infection of susceptible cells:
  - *SARS-CoV2 spike (S) proteins adhere to the membrane of cells (ACE-2 receptors) and fuse with the membrane.*
  - *Viral RNA is translated into viral proteins in the rhibosomes.*
  - *This includes RNA-dependent RNA polymerase, that produces more viral RNA.*
  - *Viral proteins form new viruses that can exit the cells.*
  - *RNA is degraded in the cytoplasm.*

- Infection of susceptible (alveolar) cells leads to increased viral load.
- Infection of susceptible cells leads to increase in pro-inflammation signaling (innate and adaptive immuno response – antibodies to recognise antigens).
- Infection of susceptible cells leads to increase in tissue damage.
- Tissue damage increases pro-inflammation.
- Free viral load leads to increase in pro-inflammation signaling.
- Pro-inflammation signaling reduces viral load.
- Pro-inflammation signaling reduces infection of susceptible cells.
- Pro-inflammation signaling increases tissue damage through induced cell death.
- Pro-inflammation signaling increases anti-inflammatory signaling (T-cells, interleukins, TGF-Beta).
- Anti-inflammatory signaling reduces pro-inflammatory signaling.



Model B: pharmacological model

- **Problem statement**: considering the pharmacological effect of infectisivir®/ Modern Phaser® vaccine. What formulation (drug delivery) effects and patient characterstics should be considered?
- **Task**: Based on the available information, develop a conceptual pharmacological model (causal loop diagram/stocks-and-flows).
- **Question 1:** Given the structure of the model, what can we say about the above question?
- **Question 2:** Are there any missing aspects in the data/model?
- **Question 2:** Why would we want to translate this into a quantitative simulation model? How? (what technique)

Model B(i): pharmacological model - infectisivir

Data Table. Pharmacological mechanism of the infectisivir drug therapy.

About the drug:

- Infectisivir is an adenosine nucleotide analogue prodrug that is metabolically activated to GS-44992. The compound acts to reduce SARS-CoV-2 viral load.

Drug disposition:

- Infectisivir is administered to a person as an intravenous injection into systemic blood.
- From systemic blood, Infectisivir is distributed into peripheral tissues and back into blood.
- Infectisivir is distributed to the liver through blood flow and is metabolised by enzymes into GS-44992.
- GS-44992 will is distributed back to systemic blood from the liver and into (and back from) peripheral tissues.
- GS-44992 is eliminated from the kidneys into urine through passive filtration.

Mechanism of action:

- Inside SARS-CoV2 infected cells, GS-44992 binds to RNA-dependent RNA polymerase RDRP - a SARS-CoV-2 protein that produces more SARS-CoV-2 viral RNA (encoded in SARS-CoV2 and produced by the rhibosomes in human cells).
  GS-44992 interferes with RDRP RNA production by acting as a adenine analogue, this eventually stops the production of viral RNA through RDRP.
- As a counteracting effect, the viral protein exoribonuclease detects and fixes errors in viral RNA.
- Infectisivir then reduces viral RNA and improves lung function.



Model B(ii) – The “Modern Phaser” Vaccine

Data Table. Pharmacological mechanism of the Modern Phaser Vaccine.

About the vaccine:

- The “Modern Phaser” vaccine contains strands of RNA encoding for a SARS-CoV2 antigen, encased in lipid nanoparticles (LNP) to avoid elimination by the human immunesystem.

Vaccine disposition:

- The vaccine is administered by intramuscular injection.
- LNPs are distributed into the central blood flow, and from central blood flow.
- LNPs are distributed from central blood into peripheral tissues, and back again.
- LNPs can distribute into cells across the cellular membrane.
- LNPs degrade over time in the body.

Mechanism of action:

- LNPs release mRNA for producing spike proteins of SARS-CoV-2 (antigen), making use of the cellular functions for protein production.
- mRNA is degraded in the cytoplasm.
- Spike proteins are present on cell membrane.
- The spike proteins trigger an immune response.
- The innate immune system attempts to eliminate infected cells.
- Over a longer duration of time, the adaptive immune systems builds up memory cells (antibodies) for recognising and eliminating SARS-CoV-2.

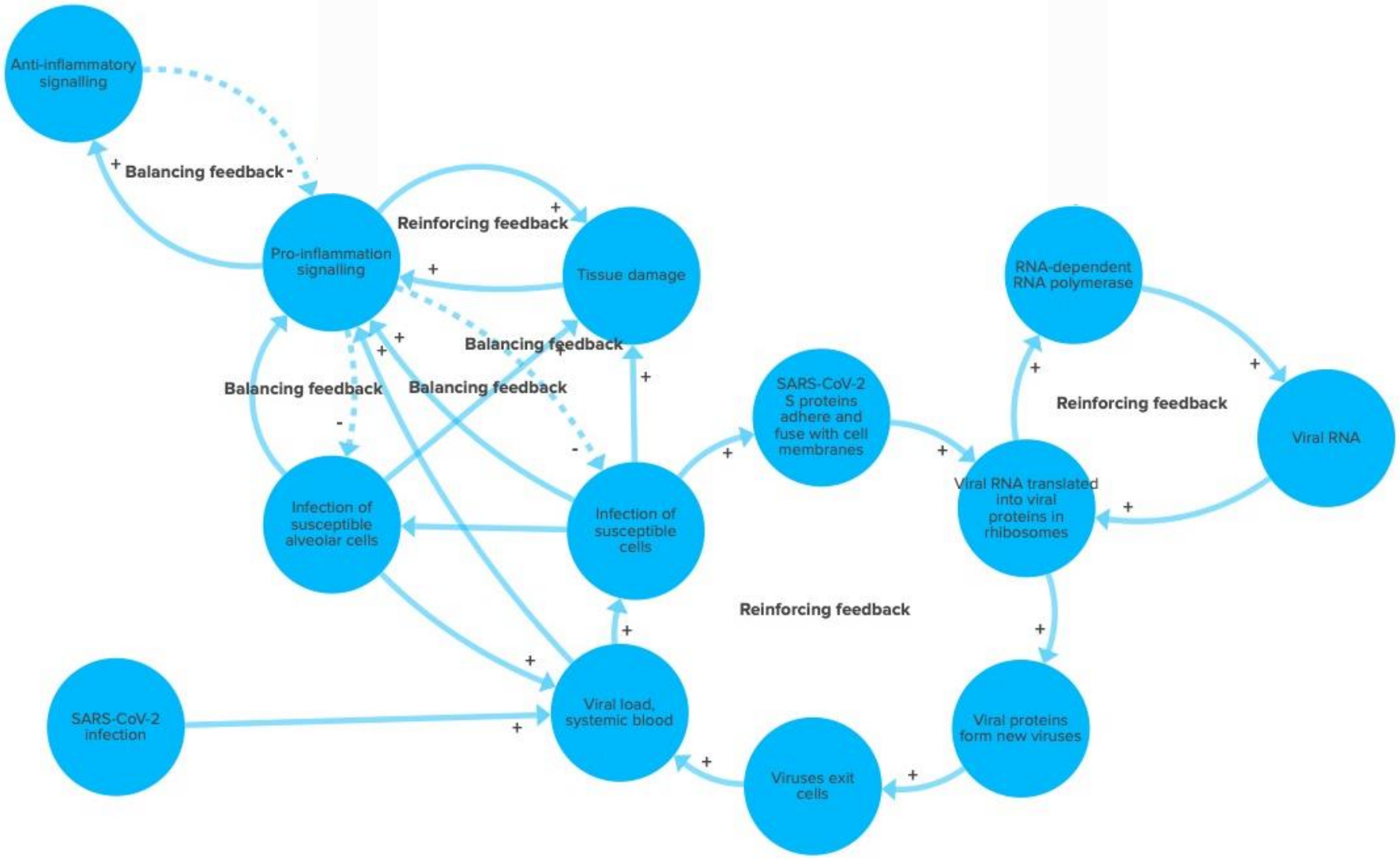


Model A&B: The combined model

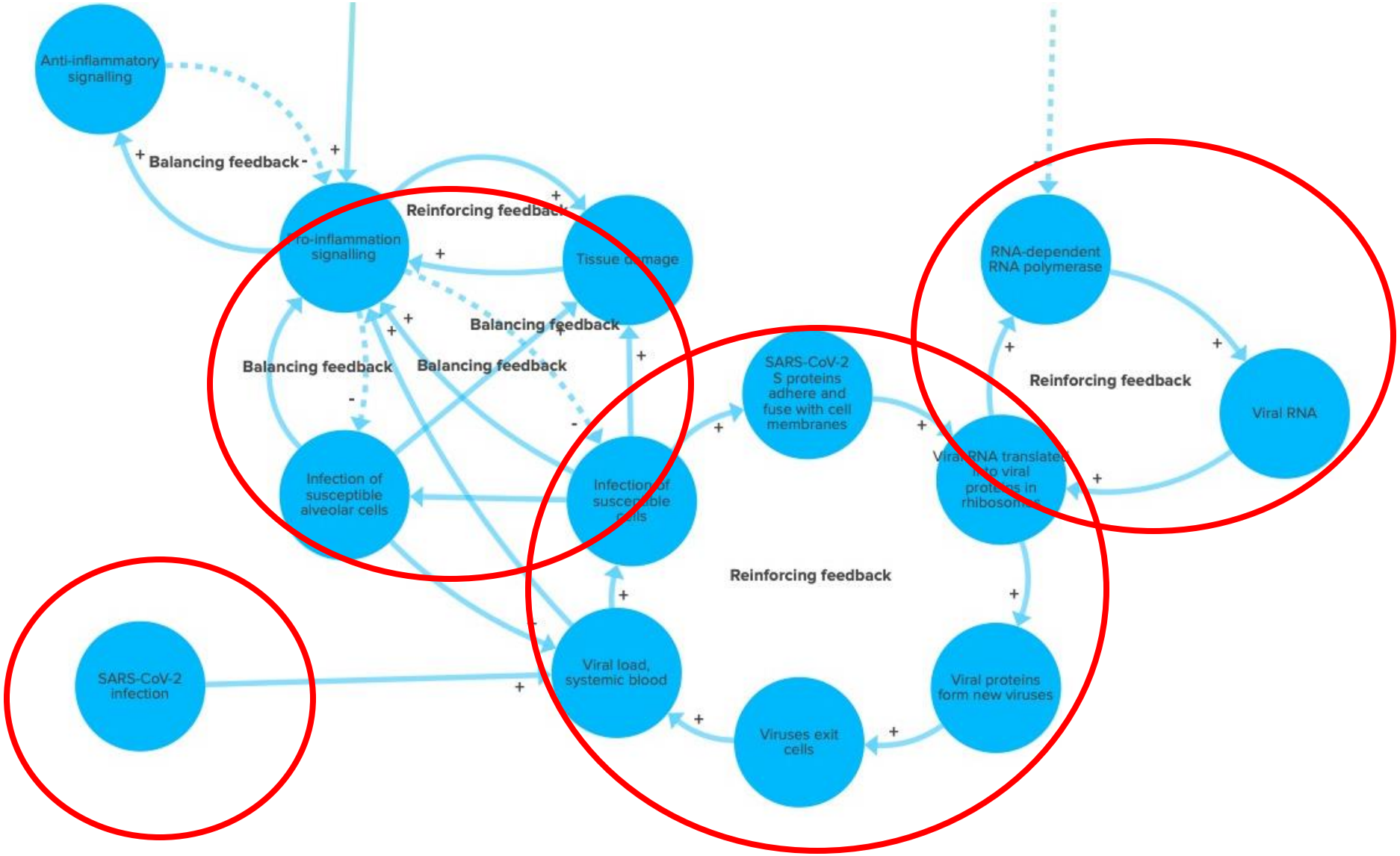
- In the larger group, discuss how these models could be combined and where the links would be in the respective models.
- What is the added value of a combined model?
- What are the disadvantages?
- And finally, what is the added value of a systems modeller (*e.g.*, medical engineer) in pharmaceutical development?

Example solutions

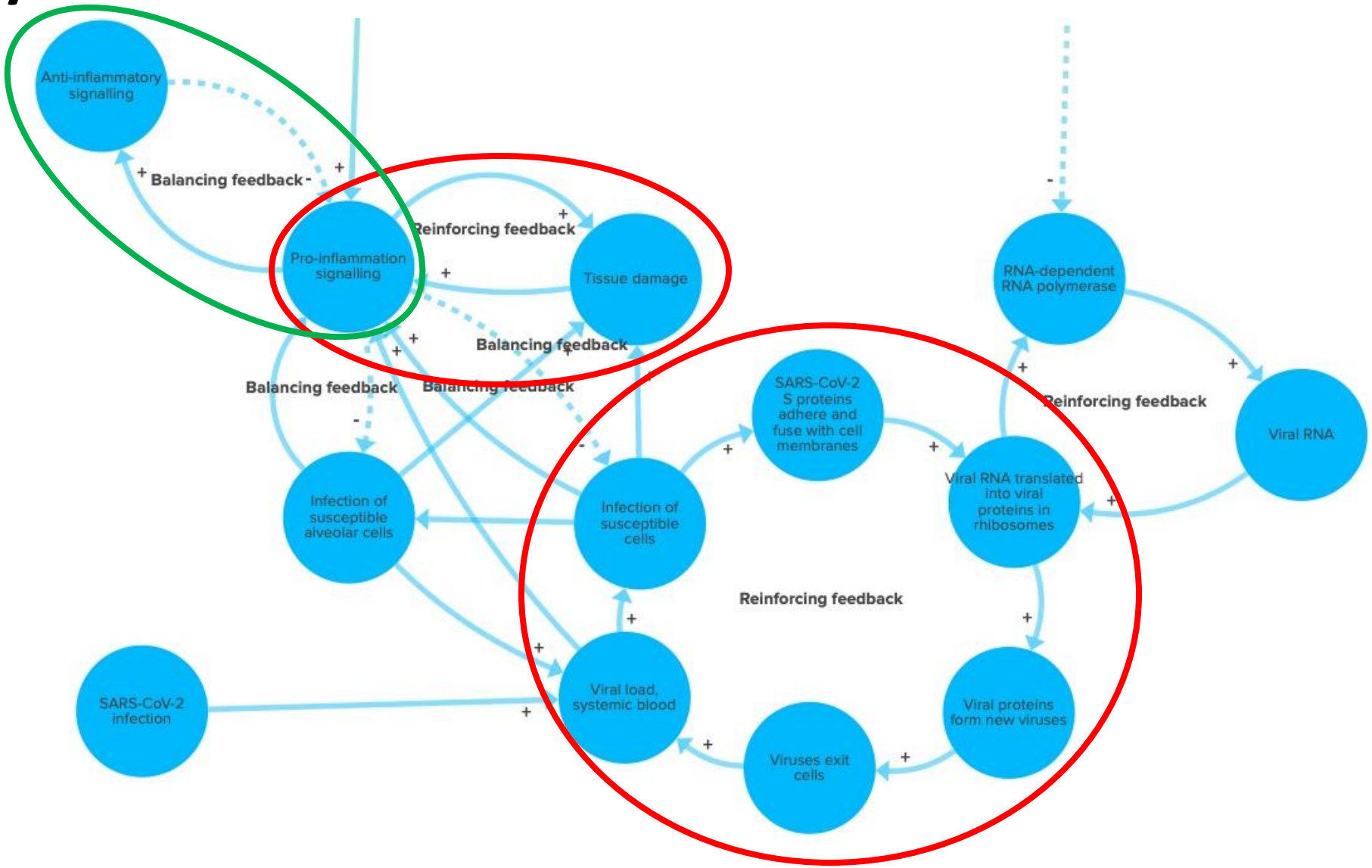
Disease model



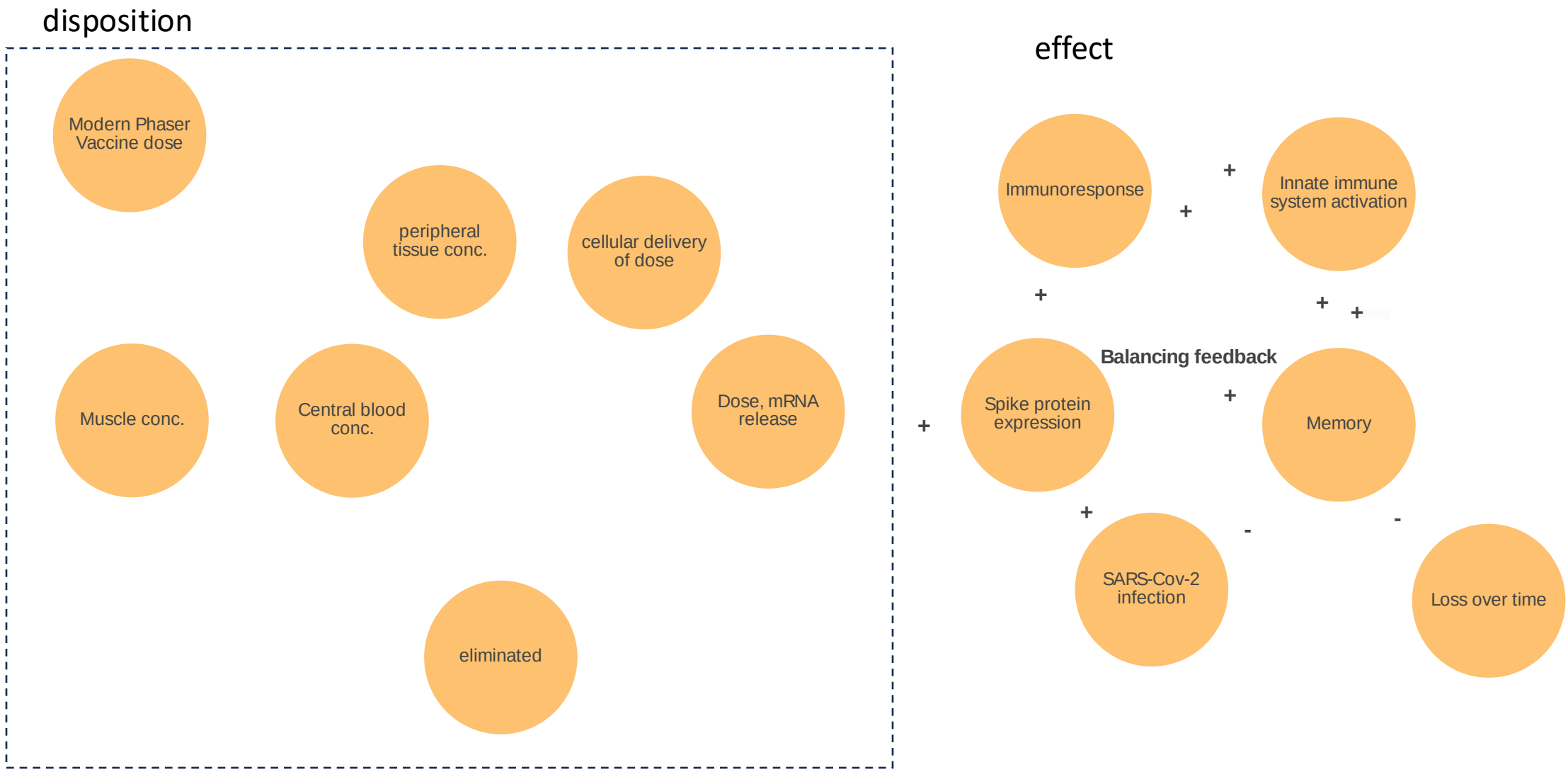
Potential drug/treatment/policy targets



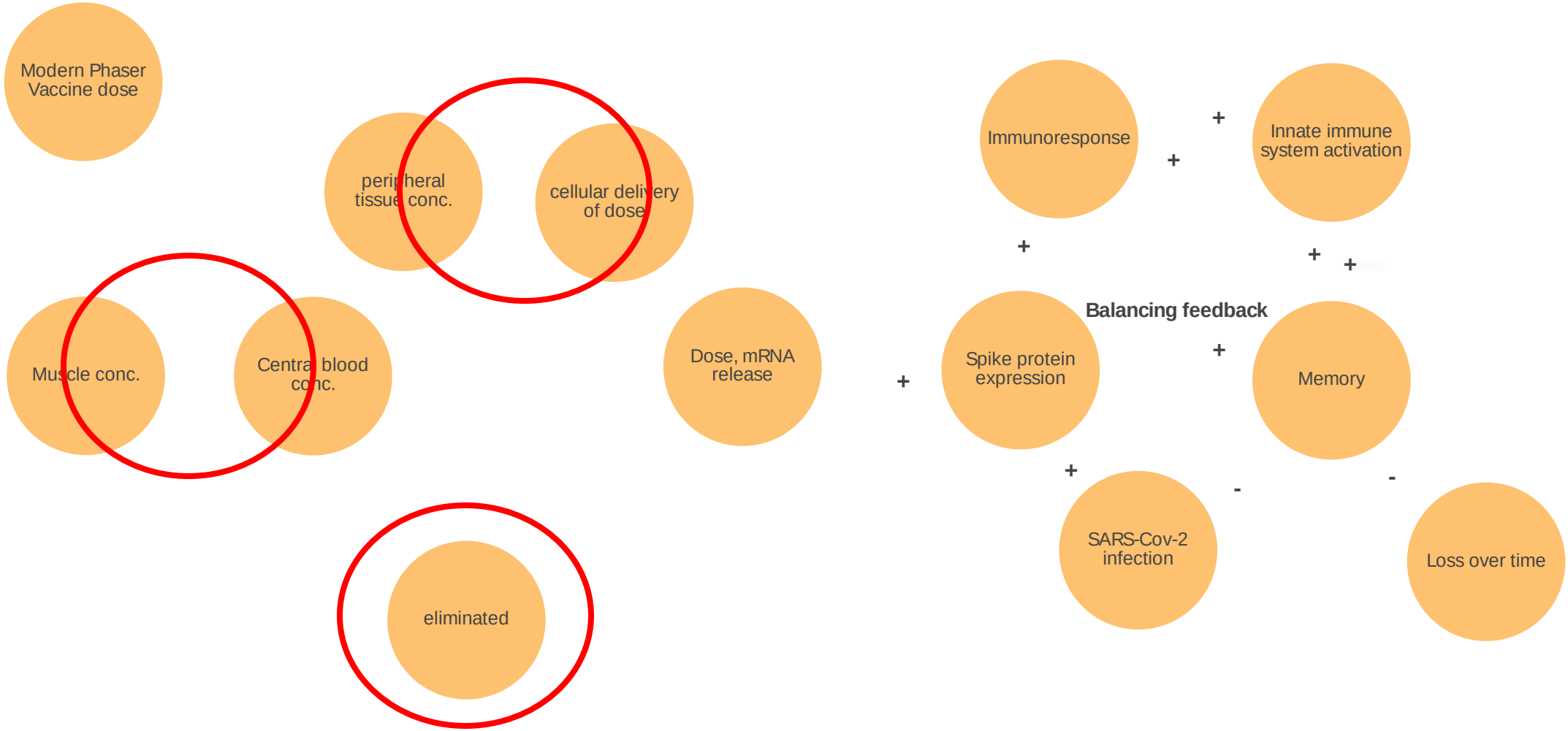
Severity of disease



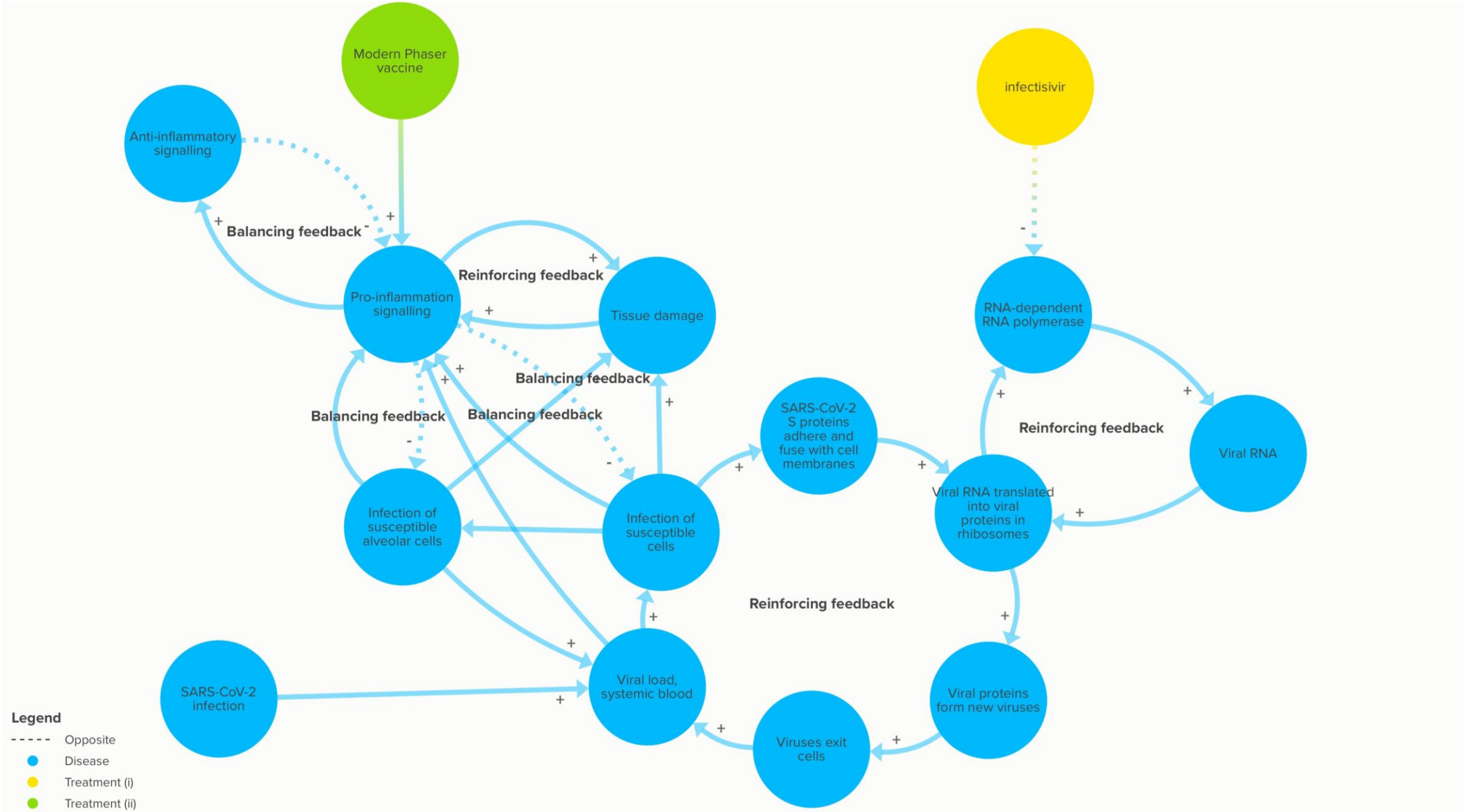
model: vaccine



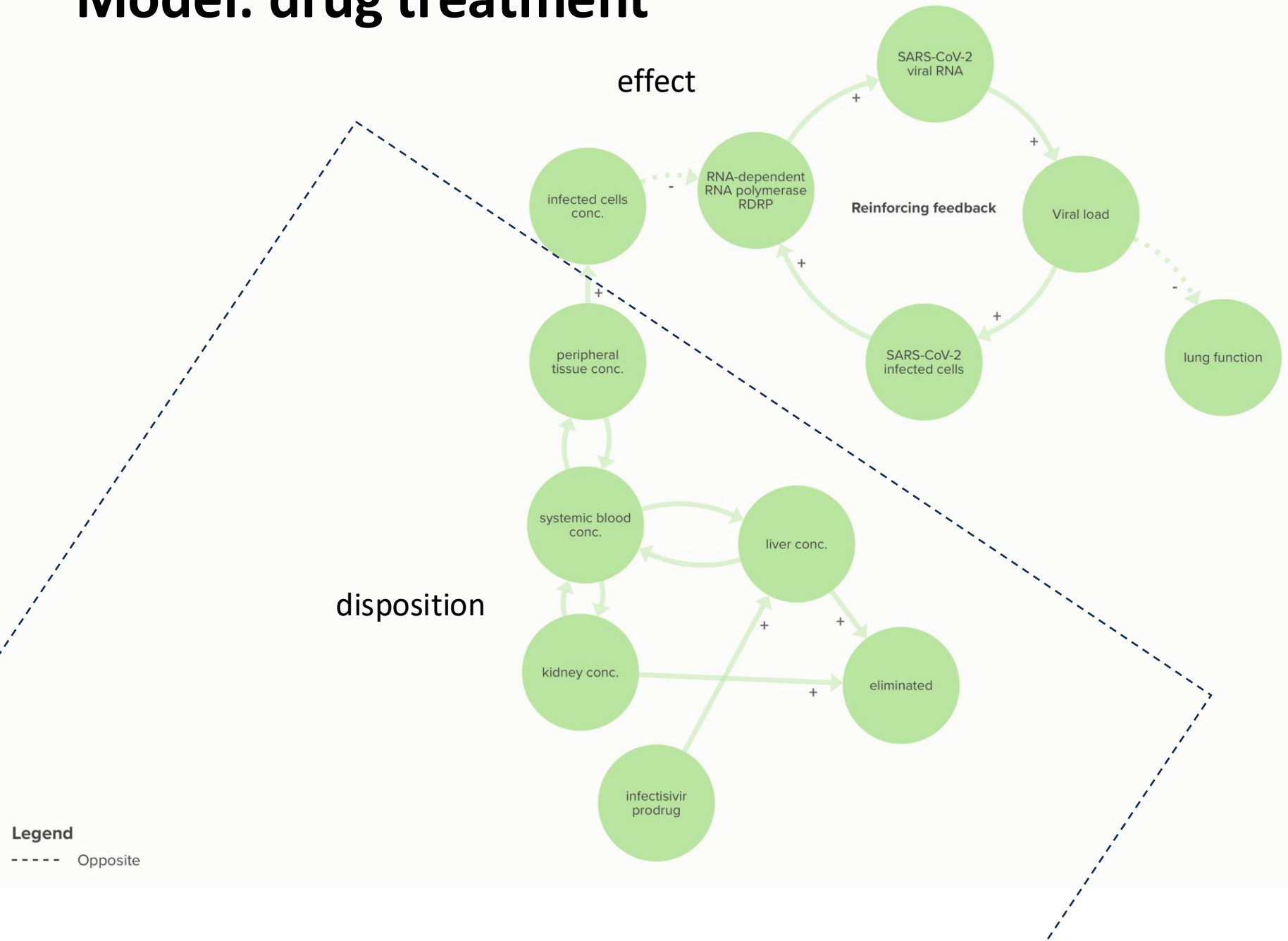
vaccine: formulation/delivery



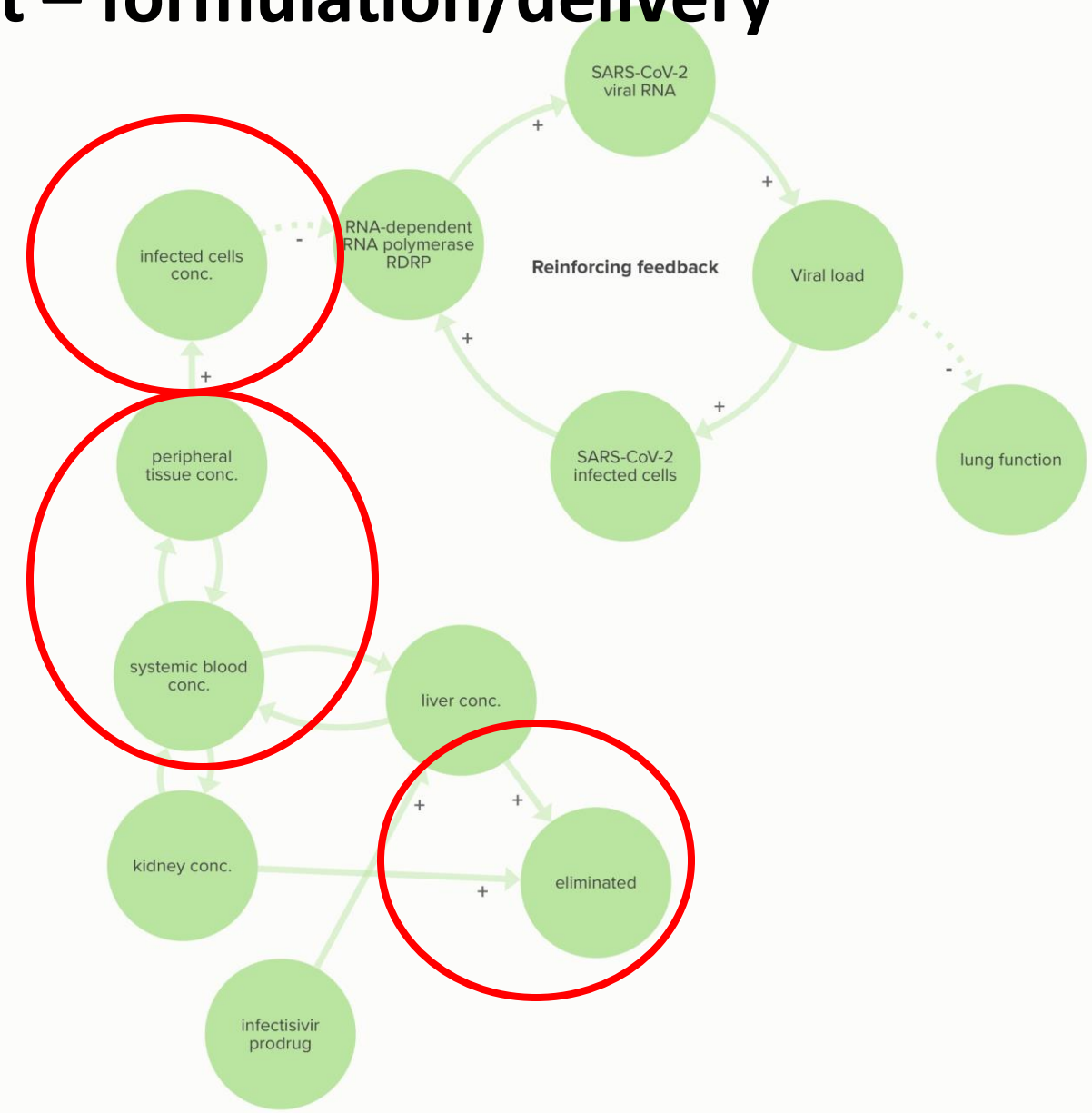
combined model



Model: drug treatment



drug treatment – formulation/delivery



Feedback

